

Chapter 4: Phylogeny and Evolutionary History

Phylogeny and Classification

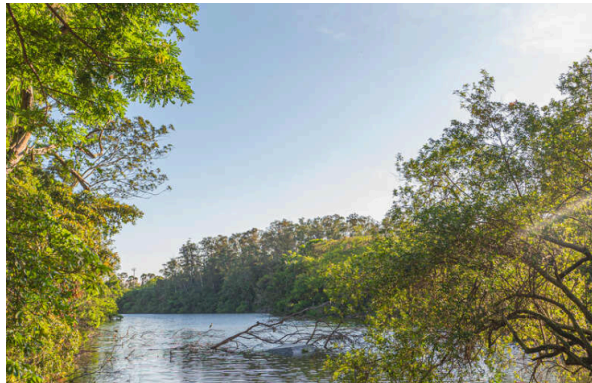
- **Biodiversity** is immense and requires systematic classification.

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- The diversity of life motivates the need for scientific classification.
- **Linnaean taxonomy** organizes species hierarchically but was not based on evolutionary theory.
- **Darwin** explained that **branching descent with modification** produces nested hierarchies.
- **Phylogenetic systematics** classifies organisms by evolutionary history, using **clades** (monophyletic groups).

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- Phylogenies can be constructed at different scales, from the tree of life to specific lineages.
- **Phylogeny** is the study of branching relationships among populations over evolutionary time.

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- Pedigrees trace individual ancestry; phylogenies trace population/species ancestry.
- **Traits** (morphological, molecular, behavioral) are used to infer phylogeny.

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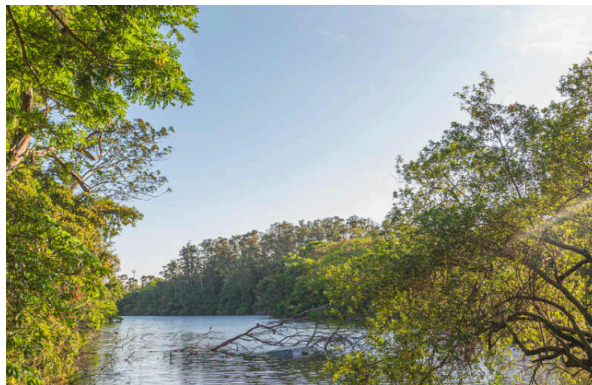


- Traits are used both to reconstruct trees and to map evolutionary events.

Phylogenies Reflect Evolutionary History

- The goal of phylogenetics is to reconstruct patterns of descent and understand evolutionary events.
- Phylogenies can be constructed for all life, for genera, species, or populations.

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- Phylogenetic trees can be drawn in tree or ladder format; orientation does not affect meaning.

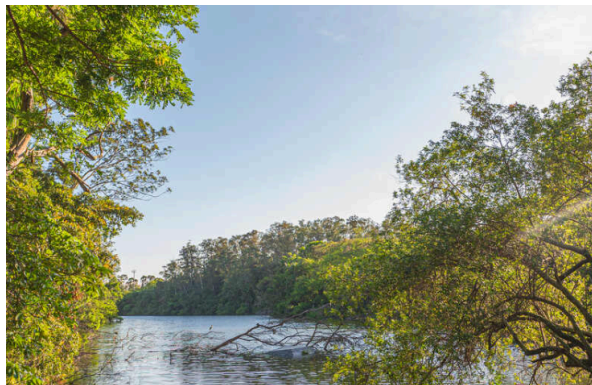
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- Interior nodes represent common ancestors; tracing back identifies the most recent common ancestor.

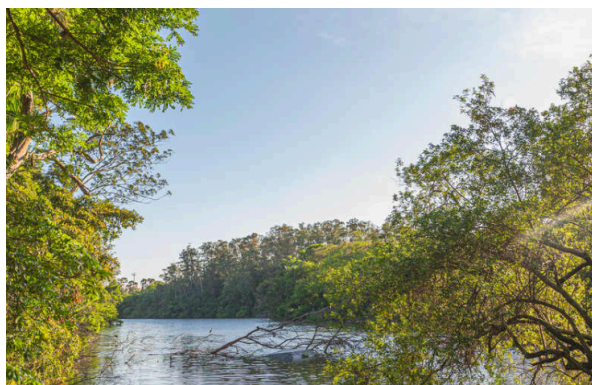
- **Node rotation** does not change relationships.

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- Rotating nodes leaves relationships unchanged.

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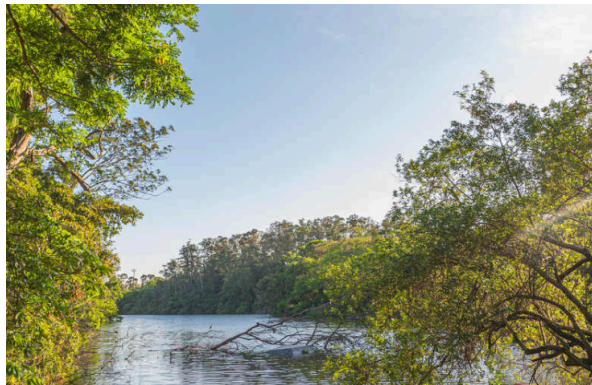
- Multiple layouts can represent the same relationships; focus on branching patterns.
- **Clades** are monophyletic groups (all descendants of a common ancestor).

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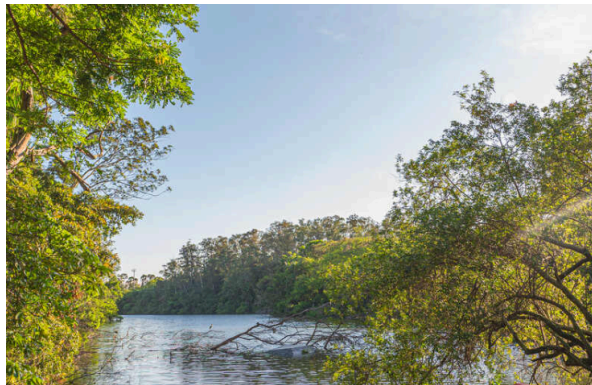
- Polytomies represent uncertainty in relationships.

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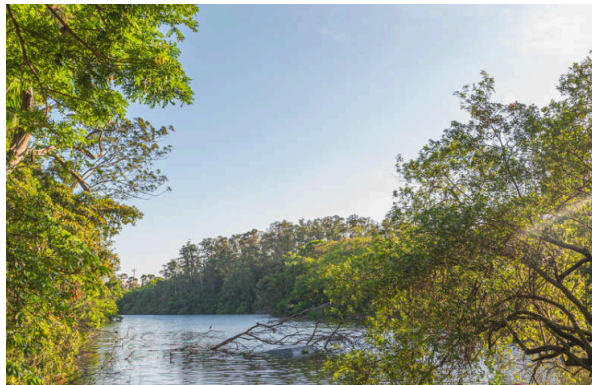
- Clades are nested; each clade includes all descendants of a single ancestor.

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- Monophyletic, paraphyletic, and polyphyletic groups: only monophyletic groups are true clades.

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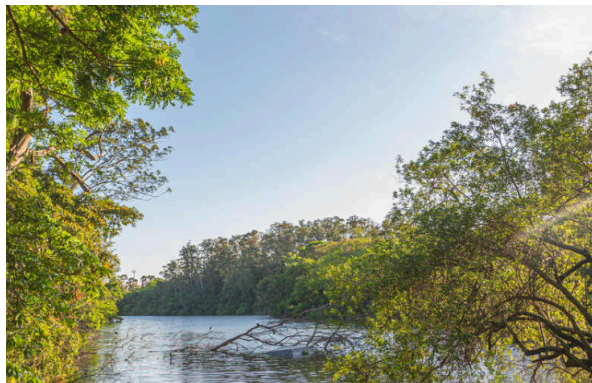
- Paraphyletic groups (e.g., “fish”) exclude some descendants; only clades reflect true history.
- **Rooted trees** indicate time direction; **unrooted trees** show relationships without time.
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- Unrooted trees are common in molecular phylogenetics.
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- Root placement affects which groups are monophyletic.
- **Branch lengths** in phylograms represent evolutionary change; in cladograms, only relationships.
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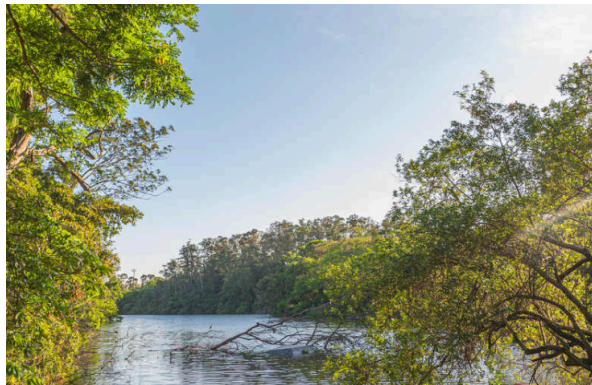
- Cladograms and phylograms: branch lengths can show amount of change.

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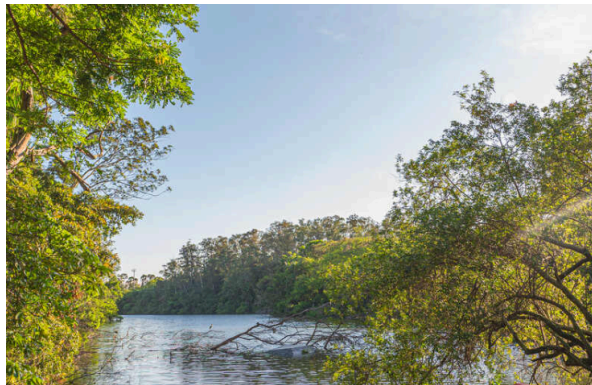
- Chronograms use branch lengths to represent time, allowing divergence time estimation.

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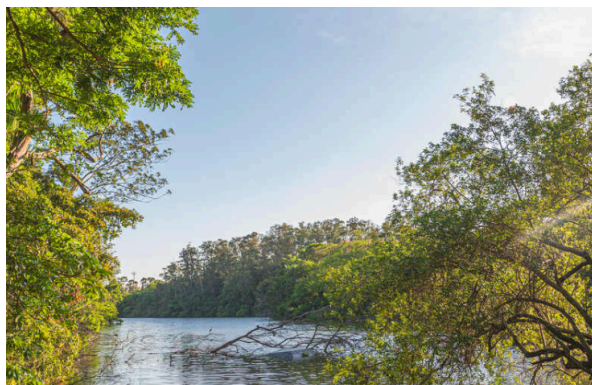
- Comparing branch lengths tests hypotheses about evolutionary rates.

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- Phylogenies can be depicted as ladders, trees, or circles.

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- Quantitative trait data (e.g., opsin sensitivity) can be mapped onto trees.

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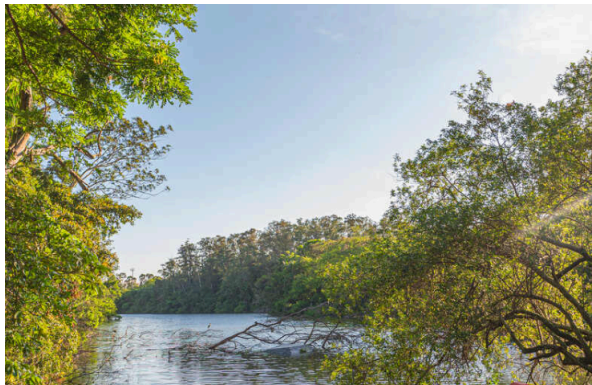


- Mapping trait gains/losses reveals the sequence and timing of evolutionary innovations.

Traits on Trees

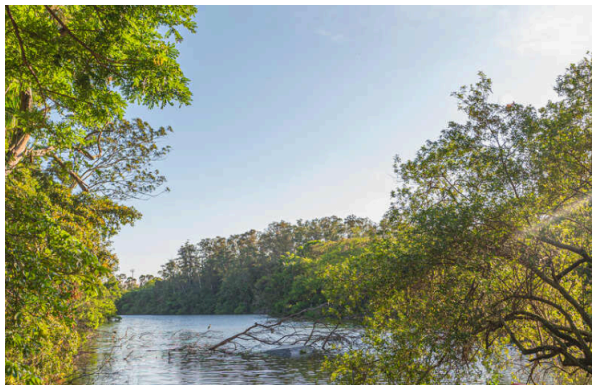
- **Character state** is the specific form of a trait used in phylogenetic analysis.
- **Homology** is similarity due to shared ancestry; **analogy** is similarity due to convergence.

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- Distinguishing homologous from analogous traits is critical for accurate phylogenies.
- **Divergent evolution** explains differences among related groups; **convergent evolution** explains similarities among unrelated groups.

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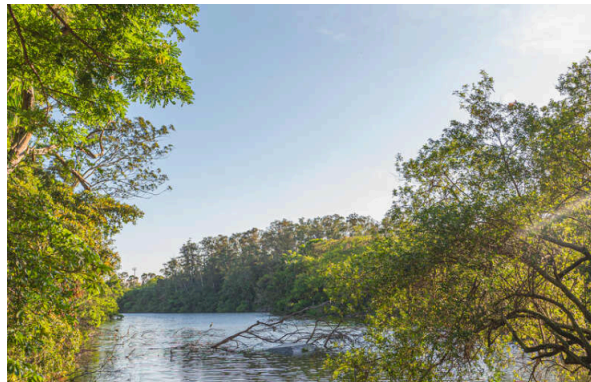
- Convergent evolution: similar adaptations in unrelated taxa due to similar environments.

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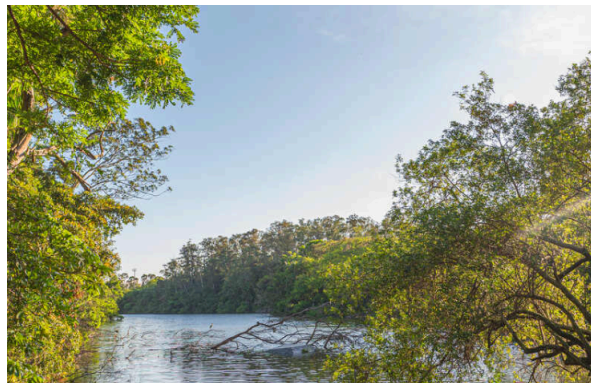
- Convergent evolution in body form (e.g., sharks and tuna) highlights the need for multiple traits.
- **Synapomorphy** is a shared derived trait defining a clade.

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- Synapomorphies group taxa by shared evolutionary innovations.
- **Homoplasy** is similarity not due to common ancestry (e.g., convergence, reversal).

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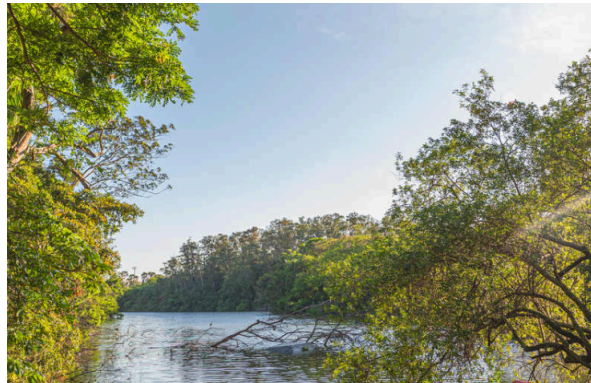
- Homoplasy can mislead tree reconstruction.
- **Symplesiomorphy** is a shared ancestral trait, not useful for recent relationships.

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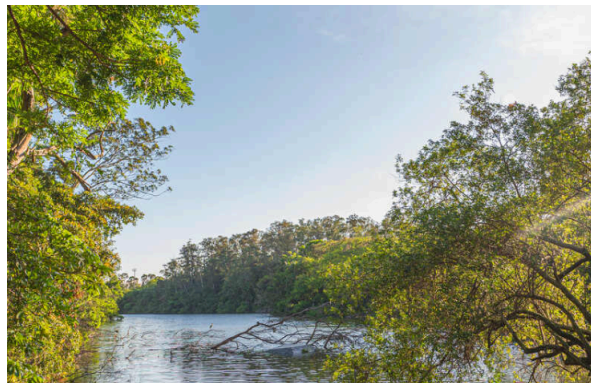
- Avoiding symplesiomorphies ensures only informative traits are used.
- **Parsimony informative character** has at least two states, each in at least two taxa.
- **Outgroup** is a taxon outside the group of interest, used to infer trait polarity and root the tree.

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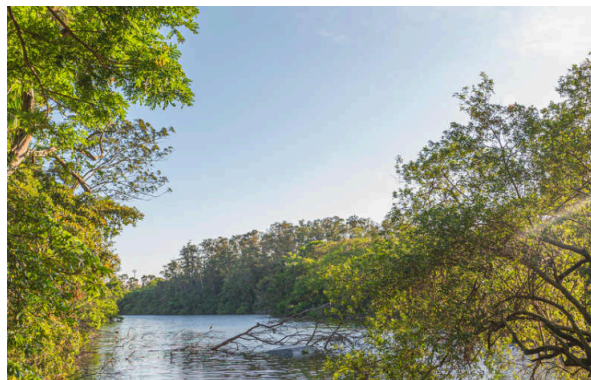
- Outgroups establish evolutionary direction and test trait evolution hypotheses.

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- Outgroups help resolve polytomies by providing a reference for the ancestral state.

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- Sometimes outgroups do not resolve all ambiguities.

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- Synapomorphies at multiple levels resolve complex histories.

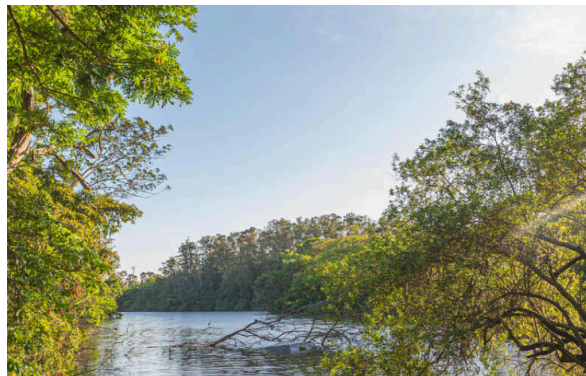
Mathematical and Statistical Methods in Phylogenetics

- **Parsimony** prefers the tree with the minimum number of changes. Score: $S = \sum_{\{i=1\}}^n c_i$ where c_i is the number of changes for character i .
- **Branch length** in a phylogram: rate of evolution $r = \frac{d}{t}$ where r is rate, d is genetic distance, t is time.
- **Divergence time estimation** (molecular clock): $t = \frac{d}{r}$ where t is time, d is genetic distance, r is rate.
- **Likelihood model**: The likelihood of a tree T given data D : $L(T) = P(D | T, M)$ where M is the model of evolution.
- **Statistical support** for trees is assessed using bootstrap values or posterior probabilities.

Using Phylogenies to Generate Evolutionary Hypotheses

- Phylogenies are testable scientific hypotheses about the history of life.
- Mapping traits onto trees allows inference of the sequence and timing of evolutionary events.

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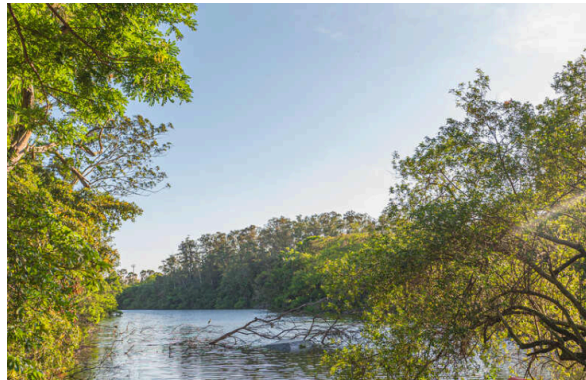
- The evolution of complex traits (e.g., venom delivery) can be traced using phylogenies.

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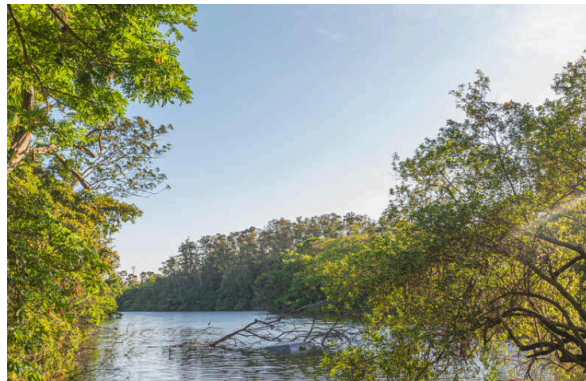
- Phylogenetic analysis can reveal widespread homology of traits once thought unique.

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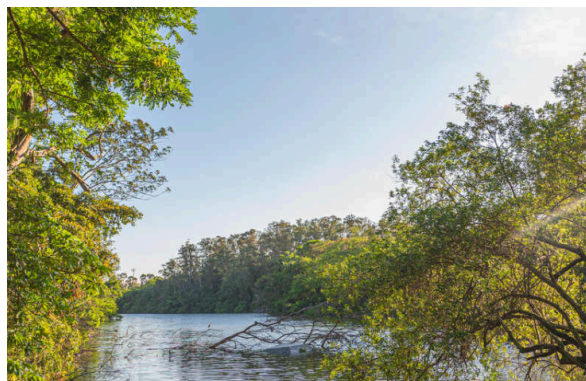
- Shared toxin genes in snakes and lizards support a single evolutionary origin for venom.
- **Deep homology** is the conservation of genetic/developmental mechanisms across distant taxa.

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- Lipid droplet formation is deeply conserved, illustrating deep homology.

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- Molecular phylogenies based on conserved genes confirm evolutionary relationships.

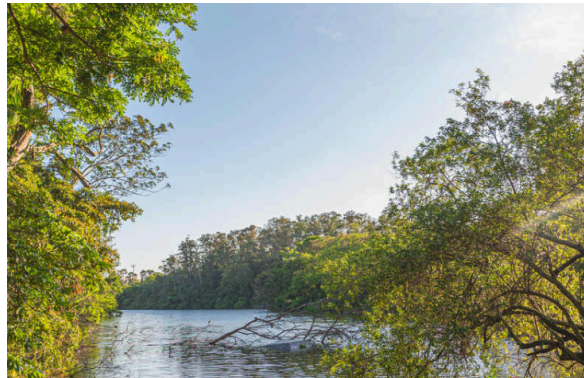
- **Vestigial trait** is a reduced or functionless structure inherited from an ancestor.

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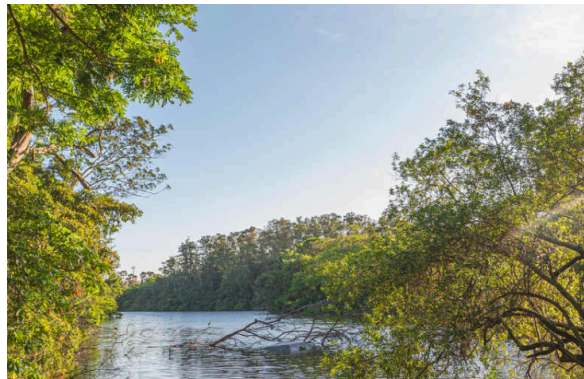
- Vestigial structures (e.g., plica semilunaris) support common ancestry.

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- Limb loss in snakes, documented by vestigial structures and fossils, exemplifies evolutionary modification.

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- The distribution of vestigial traits matches evolutionary predictions.

Key Terms

- Phylogeny, clade, monophyletic group, paraphyletic group, polyphyletic group, homologous trait, analogous trait, synapomorphy, homoplasy, symplesiomorphy, outgroup, rooted tree, unrooted tree, cladogram, phylogram, vestigial trait, divergent evolution, convergent evolution, polarity, node, taxon.